Proportions from samples

1 A television station wants to estimate the number of viewers it had for a new show. They know that the country's population is $13\,620\,000$. When they randomly selected 5000 people and asked them if they had watched the show, they found that 400 of them said 'Yes'. Estimate the number of people who watched the show in the entire population.

2 The following table shows the gender of fifty Grade 12 students at a particular school:

Students 1 - 10	M, F, M, F, F, M, M, M, F, M
Students 11 - 20	M, F, M, M, F, M, F, M, F, F
Students 21 - 30	F, F, M, M, F, M, M, F, F, F
Students 31 - 40	M, F, M, F, F, F, M, M, M, F
Students 41 - 50	F, M, F, M, M, F, M, F, M, F

a Calculate the proportion of females in the sample.

b What proportion of the first 5 students were female?

c What proportion of the first 10 students were female?

d What proportion of the first 20 students were female?

e In a systematic sample, every second student is chosen, in the order that they appear, from the first 20 students. How many males will be chosen in the sample?

f In a systematic sample, every third student is chosen, in the order that they appear, from the first 40 students. How many females will be chosen in the sample?

g The school has a population of 440 students. If the proportion of males and females in the sample is indicative of the whole school, how many female students are there in the school?

3 An oil spill has spread over an area of 1650 km^2. A team of marine biologists scan an area of 150 km^2, and find 272 dead marine animals.
Find an estimate for the number of dead marine animals over the entire area of the oil spill.

Population and sample mean

4 The heights (in cm) of a population of three people are represented by A, B and C.

a List all possible samples of size two without replacement. For example if the first two are selected we can write that as AB.

b If $A = 171$, $B = 153$ and $C = 162$, complete the following table:

Sample values (cm)	Sample mean (cm)
$171, 153$	
$171, 162$	
$153, 162$	

c What is the mean of the distribution of all possible sample means?

d What is the population mean?

e Is the mean of the sample means equal to the population mean?

5 The weights (in kg) of a population of five people are represented by F, G, H, I and J.

a List all possible samples of size four without replacement. For example if the first four are selected we can write that as $FGHI$.

b If $F = 96$, $G = 120$, $H = 46$, $I = 56$ and $J = 89$, complete the following table:

Sample values (kg)	Sample mean (kg)
$96, 120, 46, 56$	
$96, 120, 46, 89$	
$96, 120, 56, 89$	
$96, 46, 56, 89$	
$120, 46, 56, 89$	

c What is the mean of the distribution of all possible sample means?

d What is the population mean?

e Is the mean of the sample means equal to the population mean?

6 The weights (in kg) of a population of five p[redacted] are represented by F, G, H, I and J.

a List all possible samples of size one without replacement.

b If $F=98$, $G=136$, $H=116$, $I=94$, and $J=130$ complete the following table:

Sample values (kg)	Sample mean (kg)
98	
136	
116	
94	
130	

c What is the mean of the distribution of all possible sample means?

d What is the population mean?

e Is the mean of the sample means equal to the population mean?

Capture-recapture

7 Marine biologists want to determine the number of dolphins left along a stretch of coast. They tag 72 dolphins. After a couple of years they come back to the same area and check 84 dolphins, 12 of which have tags.

a Find the proportion of dolphins that had tags upon checking.

b Find x, the expected population of dolphins along that coast.

8 Conservationists previously captured, tagged and released 400 whales in a sanctuary. Some time later a large number of whales were observed, and 34% were found to be tagged. Estimate n, the population of the whales in the sanctuary.

9 A local council wanted to monitor the number of rabbits in the area. They used the capture-recapture technique to estimate the population of rabbits. 219 rabbits were caught, tagged and released. Later, 42 rabbits were caught at random. 15 of these 42 rabbits had been tagged.

a Find the estimated population of rabbits. Round your answer to the nearest whole number.

b Local council B conducted a similar study and found they had 15% fewer rabbits. What was the estimated population of rabbits in local council B? Round your answer to the

10 A researcher captures a sample of 400 fish in a river, tags them and then releases them. The following day, he captures a sample of 1200 fish, of which 100 have tags.
Estimate the number of fish in the river.

11 A gamekeeper captures a sample of 600 animals, comprised of 100 deer, 300 lions and 200 zebras, tags them and then releases them. The following day, he captures a sample of 200 deer of which 20 are tagged, 1200 lions of which 75 are tagged, and 400 zebras of which 50 are tagged.
Estimate the number of the following animals in the game reserve:

a Deer **b** Lions **c** Zebras

12 A researcher captures a sample of 300 fish in a river, tags them and then releases them.

a The following day, he captures a sample of 450 fish, of which 75 have tags attached to them. Estimate the number of fish in the river.

b The researcher releases back the first sample. The day after that, he captures a sample of 750 fish, of which 30 have tags attached to them. Estimate the number of fish in the river using the results of this sample.

c Find the average of the two estimates.

13 The capture-recapture technique was used to estimate the population of cuttlefish. At the beginning of the year, the number of cuttlefish caught, tagged and released was 210. When 700 cuttlefish were taken at the end of the year, 140 of them were found to be recaptured tagged ones.

a What percentage of the cuttlefish captured at the end of the year were tagged?

b Find the estimated population of cuttlefish in the river.